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Validity and Reliability of a Motorized Sprint Resistance Device

Elvir Rakovic,[1] Gøran Paulsen,[2,3] Christian Helland,[2] Thomas Haugen,[4] and Ola Eriksrud[3]

[1]Department of Food and Nutrition and Sport Science, University of Gothenburg, Gothenburg, Sweden; [2]Norwegian Olympic and Paralympic Committee and Confederation of Sports, Oslo, Norway; [3]Department of Physical Performance, Norwegian School of Sport of Sciences, Oslo, Norway; and [4]School of Health Sciences, Kristiania University College, Oslo, Norway

Abstract

Rakovic, E, Paulsen, G, Helland, C, Haugen, T, and Eriksrud, O. Validity and reliability of a motorized sprint resistance device. *J Strength Cond Res* 36(8): 2335–2338, 2022—An increasing number of sprint-related studies have used motorized devices to provide resistance while sprinting. The aim of this study was to establish within-session reliability and criterion validity of sprint times obtained from a motorized resistance device. Seventeen elite, female, handball players (22.9 ± 3.0 years; 176.5 ± 6.5 cm; 72.7 ± 5.5 kg; training volume 9.3 ± 0.7 hours per week) performed two 30-m sprints under 3 different resistance loading conditions (50, 80 and 110 N). Sprint times ($t_{0–5m}$, $t_{5–10m}$, $t_{10–15m}$, $t_{15–20m}$, $t_{20–30m}$, and $t_{0–30m}$) were assessed simultaneously by a 1080 Sprint motorized resistance device and a postprocessing timing system. The results showed that 1080 Sprint timing was equivalent to the postprocessing timing system within the limits of precision (±0.01 seconds). A systematic bias of approximately 0.34 ± 0.01 seconds was observed for $t_{0–5m}$ caused by different athlete location and velocity at triggering point between the systems. Coefficient of variation was approximately 2% for $t_{0–5}$ and approximately 1% for the other time intervals, although standard error of measurement ranged from 0.01 to 0.05 seconds, depending on distance and phase of sprint. Intraclass correlation ranged from 0.86 to 0.95. In conclusion, the present study shows that the 1080 Sprint is valid and reliable for sprint performance monitoring purposes.

Key Words: spatiotemporal measurements, sprint conditioning, photocells, resisted sprinting

Introduction

Sprint training and testing are common routines for many athletes and coaches. Such practices are accompanied by a variety of modalities (e.g., linear or change-of-direction sprints, accelerated or maximal velocity sprinting), loading components (duration, intensity, resting periods, session rate, resisted/assisted conditions, etc.), procedures (e.g., time initiation and starting position), and equipment (timing gates, laser guns and radar devices, Global Positioning System, sleds, towing cords, footwear, etc.) (3,4).

An increasing number of sprint-related studies have used motorized devices to provide resistance while sprinting, with the 1080 Sprint (1080 Motion AB, Stockholm, Sweden) commonly used (1,7–9,11). Application of such a device may serve several benefits. First, an accurate resistance can be predetermined, which is more challenging with, for example, sleds due to surface friction issues under varying environmental conditions. Moreover, synchronized assessments of velocity and displacement relative to the start line with the force exerted through the machine's cord under varying loading conditions can be obtained by one device only. This will negate the need for the combination of sleds with photocells, laser guns, or radars. The distance-time or velocity-time running data can in turn be used for the computation of macroscopic mechanical outputs (10) that may form basis for individual training prescription (1,9,10). However, these potential benefits are dependent on the ability of the motorized device to accurately assess velocity-time data. To the best of our knowledge, no studies to date have addressed this issue. The purpose of this study was therefore to determine within-session reliability and criterion validity of sprint split times obtained from a 1080 Sprint motorized device.

Address correspondence to Ola Eriksrud, ola.eriksrud@nih.no.

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Methods

Experimental Approach to the Problem

The data used for this reliability and validation study were compiled from anonymized data from a previously published investigation exploring the effect of individual sprint training prescription based on force-velocity (FV) profiles (9). Because it is crucial that the entire acceleration phase of sprinting athletes is covered by timing gates to ensure valid and reliable FV profiles (10), the female elite team sport athletes performed 30-m sprints with varying resistance loading. Split times ($t_{0–5m}$, $t_{5–10m}$, $t_{10–15m}$, $t_{15–20m}$, $t_{20–30m}$, and $t_{0–30m}$) were assessed simultaneously by a motorized resistance device and a postprocessing timing system. These measurements formed basis for intrasession reliability and validity assessments.

Subjects

Seventeen elite, female, handball players (mean ± *SD* [age range: 19–30]: 22.9 ± 3.0 years; 176.5 ± 6.5 cm; 72.7 ± 5.5 kg; total training volume: 9.3 ± 0.7 hours per week) with a minimum of 10-year handball-specific conditioning volunteered to participate. Four of these played for the national team, whereas 11 players participated in the Champions League tournament during the current season. The study was reviewed by the Regional Ethics Committee and approved by the Norwegian Data Protection Authority. Because of the newly implemented General Data Protection Regulations (GDPR) by the European Union, the local university Norwegian School of Sport Sciences has the responsibility for data security and ethics. All subjects signed a written informed consent form before participation, and this study was conducted according to the Declaration of Helsinki.

Procedures

A standardized 20-minute warm-up consisting of jogging (~60–75% of age-predicted maximal heart rate), selected exercises (lunges, hip lift, ballistic mobility hamstrings and hips in prone and supine), running drills (high knees, skipping, butt-kicks, straight leg pulls), and 3 to 4 sprints with increasing speed was conducted before testing (9). After the warm-up, the athletes performed 2 maximal 30-m sprints with 50, 80, and 110 N resistance, respectively, in a randomized order (i.e., one sprint with each resistance before proceeding to the next sequence). The resistance during the 6 sprints was provided by a 1080 Sprint motorized device (1080 Motion AB, Stockholm, Sweden). All sprints were initiated from a standing, split-stance position with the tip of the toe of the front foot placed on the start line. All starts were commenced from a static position, meaning that "leaning backward before rolling forward" was not allowed. After a ready signal was given by the test leader, the athletes started on their own initiative. Recovery time between each sprint was approximately 4 minutes.

MuscleLab timing system (Ergotest AS, Porsgrunn, Norway) was used to assess sprint times. An infrared optical contact mat covered the start line, and timing was initiated at the point of front foot lift-off. Postprocessing timing gates (i.e., an internal software scans all signals from the timing gate in terms of frequency and duration) where mounted on tripods 120 cm above floor level and placed at 5, 10, 15, 20, and 30 m. Thus, all timing gates were mounted above hip height to avoid undue beam break caused by the lower limbs (3). The onset of the longest break of the infrared beam was used as a trigger criterion because the torso will produce a longer break than an arm (3). Earp and Newton (2) reported that the signal processing technology completely removed all false signals (i.e., time triggering caused by swinging limbs). Moreover, Rakovic et al. (9) reported excellent reliability values for this system setup because typical error (TE) and coefficient of variation (CV) were 0.03 seconds and 1.0% for 0- to 30-m sprint time and 0.08 $m \cdot s^{-1}$ and 1.4% for V_0. Hence, the MuscleLab timing system was used as gold standard for sprint performance assessments in this study.

The 1080 Sprint was used to provide resistance and assess sprint times. This portable system uses a servo motor (2000 RPM OMRON G5 Series Motor, OMRON Corporation, Kyoto, Japan) to provide resistance while sprinting. The motorized device was placed 5 m behind the starting line with the line attached to the athlete by a centrally located ring (sacrum) on a belt firmly tightened around the pelvis. The resistance load (50, 80, or 110 N) was determined and controlled by the computer application (1080 Motion, Lidingö, Sweden). The isotonic resistance mode was used because different modes are offered by the 1080 Sprint. Position trigger criterion for time initiation was set to 30 cm of line being pulled away from the machine. This corresponds to the position of the pelvis being approximately 30 cm past the start line. Data (force, position, and time) were recorded at 333 Hz.

Statistical Analyses

Mean and *SD* are presented for all sprint times. Shapiro-Wilk test was used to test the assumption of normality for each set of sprint time data, and z-scores were calculated and analyzed for both skewness and kurtosis. Intraclass correlation coefficient, standard error of measurement (*SEM*), and CV were calculated for all sprint-time intervals to determine within session reliability. Criterion validity was based on mean difference (t_{diff}), CV, and Pearson *r* correlation. Spearman rank correlation was used instead of Pearson *r*, where the data sets were not normally distributed. Bland-Altman plots were created for sprint-time difference distribution between the timing systems.

Table 1
Within-session reliability and criterion validity for 1080 Sprint.*

		Sprint times (mean ± *SD*)		Reliability			Criterion validity		
Resistance	Interval	t_{ML} (s)	t_{1080} (s)	CV (%)	*SEM* (s)	ICC	t_{diff} (s)	CV (%)	Cor.
50 N	$t_{0–5m}$	0.97 ± 0.04	1.30 ± 0.05	2.26	0.03	0.81	0.329	20.54	0.79
	$t_{5–10m}$	0.85 ± 0.03	0.85 ± 0.02	1.01	0.01	0.92	0.002	1.65	0.75
	$t_{10–15m}$	0.78 ± 0.02	0.77 ± 0.02	1.06	0.01	0.90	−0.008	2.01	0.48
	$t_{15–20m}$	0.72 ± 0.02	0.73 ± 0.02	1.03	0.01	0.92	0.007	1.60	0.73
	$t_{20–30m}$	1.43 ± 0.04	1.43 ± 0.04	0.82	0.01	0.95	0.001	0.74	0.94
	$t_{0–30m}$	4.75 ± 0.11	5.06 ± 0.12	0.91	0.05	0.93	0.308	4.49	0.94
80 N	$t_{0–5m}$	1.03 ± 0.04	1.36 ± 0.05	1.93	0.02	0.87	0.331	19.76	0.57
	$t_{5–10m}$	0.90 ± 0.03	0.91 ± 0.02	0.99	0.01	0.92	0.005	1.37	0.80
	$t_{10–15m}$	0.83 ± 0.02	0.82 ± 0.02	1.13	0.01	0.89	−0.007	1.56	0.71
	$t_{15–20m}$	0.78 ± 0.02	0.79 ± 0.02	1.34	0.01	0.85	0.010	1.50	0.79
	$t_{20–30m}$	1.55 ± 0.05	1.56 ± 0.04	1.32	0.02	0.89	0.005	0.76	0.93
	$t_{0–30m}$	5.08 ± 0.13	5.41 ± 0.14	1.12	0.04	0.90	0.320	5.97	0.94
110 N	$t_{0–5m}$	1.06 ± 0.05	1.41 ± 0.07	2.56	0.03	0.86	0.353	20.49	0.74
	$t_{5–10m}$	0.96 ± 0.04	0.96 ± 0.02	1.10	0.01	0.90	−0.002	1.52	0.82
	$t_{10–15m}$	0.89 ± 0.02	0.88 ± 0.02	1.01	0.01	0.92	−0.006	1.54	0.68
	$t_{15–20m}$	0.85 ± 0.03	0.86 ± 0.02	0.98	0.01	0.94	0.012	1.86	0.83
	$t_{20–30m}$	1.70 ± 0.05	1.71 ± 0.05	1.30	0.02	0.91	0.004	0.87	0.92
	$t_{0–30m}$	5.45 ± 0.15	5.79 ± 0.16	1.10	0.04	0.92	0.338	4.29	0.95

*t_{ML} = sprint times from the MuscleLab timing system, t_{1080} = sprint times from the 1080 Sprint motorized device, CV = coefficient of variation, *SEM* = standard error of measurement, ICC = intraclass correlation coefficient, t_{diff} = time difference between the analyzed systems, Cor. = Correlation (Pearson *r* or Spearman rank).

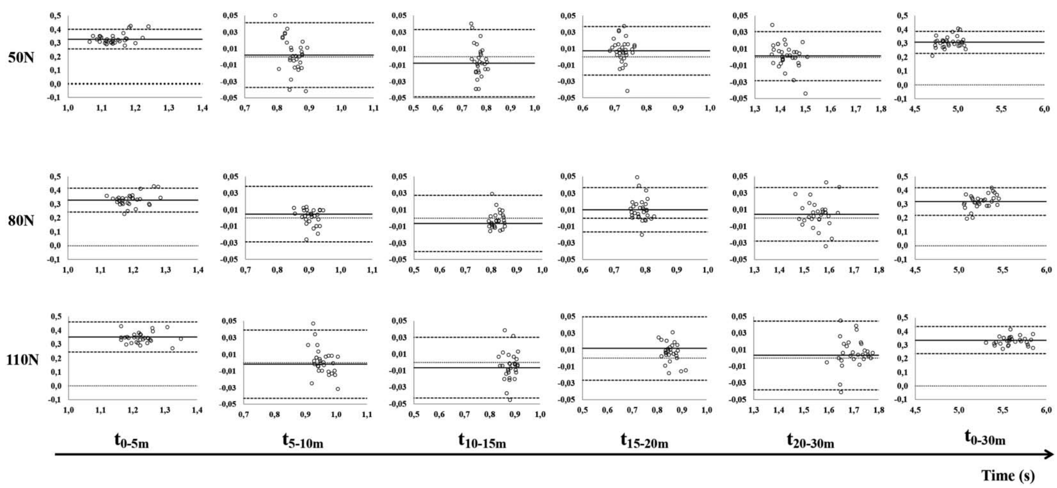

Figure 1. Bland-Altman analysis of sprint times (34 trials for each resisted condition) derived from timing gates and 1080 sprint.

Results

Table 1 shows within-session reliability and criterion validity for 1080 Sprint. Regarding reliability, CV ranged from 1.93 to 2.56% for $t_{0\text{-}5}$ and from 0.82 to 1.34 for the other time intervals, whereas *SEM* ranged from 0.01 to 0.05, depending on distance and phase of sprint. Intraclass correlation coefficient ranged from 0.86 to 0.95.

Distribution of sprint time differences for all resisted sprints are presented in Figure 1. Biases (t_{diff}) were low for $t_{5–10m}$, $t_{10–15m}$, $t_{15–20m}$, and $t_{20–30m}$ (range = −0.01 to 0.01 seconds) for all resistance conditions. Greater differences were observed for $t_{0–5m}$ (range = 0.33–0.35 seconds) and $t_{0–30m}$ (range = 0.31–0.34 seconds) across all resistance conditions.

Discussion

The aim of the present study was to explore within-session reliability and criterion validity of sprint split times obtained from a motorized device during resisted sprinting. Overall, the 1080 Sprint device displayed satisfactory reliability values. The reliability values observed are comparable to previously validated and commonly used timing systems (3). The poorest values were observed for $t_{0–5m}$. This is in line with Haugen and Buchheit (3), who reported considerably poorer reliability (TE) for $t_{0–5m}$ compared with longer sprint-distance intervals.

The current analysis revealed no systematic variation between the 1080 Sprint and the postprocessing timing gates, except for $t_{0–5m}$ and $t_{0–30m}$. That is, for practical purposes, these systems give similar results to a precision of ±0.01 seconds. Postprocessing timing gates, which were used as gold-standard in this case, are considered accurate for sprint performance monitoring because the internal software processes remove false signals (3). This provides that the timing gates are mounted above hip height (to avoid undue beam break caused by the lower limbs), as performed in this study. However, a systematic bias of approximately 0.34 ± 0.01 seconds was observed for $t_{0–5m}$ and $t_{0–30m}$. This is not surprising, as the starting method and timing system used can combine to generate large absolute differences in "sprint time" (3,6). The sources of time differences usually include the starting device, vertical and horizontal placement of starting device relative to the start line, body configuration, and velocity at triggering point (6). In this case, pelvis was approximately 60 cm past the start line at time initiation for the optical contact mat (front foot lift-off), while only approximately 30 cm past the start line at time initiation for the motorized device. Hence, pelvis was approximately 30 cm further past the start line at time initiation for the optical contact mat than for the motorized device. Provided that the bias is systematic so that correction factors can be generated (as in this case), sprint performance comparisons across systems can be performed (3). The same issue is present for the calculation of sprint mechanical outputs based on distance-time or speed-time data. An essential point when using the simple method proposed by Samozino et al. (10) is that the time 0 must be very close to the first rise of the force production onto the ground. This is equivalent to a setup with starts from blocks and audio signal with reaction time subtracted from the total time (5). According to Haugen and Buchheit (3), front foot triggering generates 0.51 seconds faster sprint times compared with starts from blocks where reaction time is subtracted from the total time. Because the current systematic bias was 0.34 seconds on average (Table 1), we estimate that a correction factor of approximately 0.17 seconds (i.e., 0.51 minus 0.34 seconds) should be added to the 1080 Sprint times to ensure valid computations of sprint mechanical outputs.

Practical Applications

The present study shows that the 1080 Sprint is valid and reliable for sprint performance monitoring purposes. This means that multiple functions for sprint training, testing, and monitoring can be operated by one device only. The benefits of using one system in both research- and field-based settings includes (a) accurate prescription of resistance while obtaining synchronized assessments of velocity, acceleration, and pulling force as a function of time or displacement relative to starting line, (b) the possibility to apply varying resistance loading during specific portions of the sprint, (c) monitor individual and team responses (i.e., fatigue), and (d) computation of sprint mechanical outputs.

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